

RESEARCH INTERESTS

Database systems: incremental view maintenance, querying unstructured and semi-structured data, and fault-tolerant distributed data systems.

EDUCATION

University of Texas at Austin Aug 2026 – Present
PhD in Computer Science
Advisors: Dixin Tang, Vijay Chidambaram

Tribhuvan University Oct 2019 – Aug 2024
BS in Computer Science and Information Technology

PUBLICATIONS

- [1] Mihai Budiu, Leonid Ryzhyk, Gerd Zellweger, Ben Pfaff, Lalith Suresh, Simon Kassing, **Abhinav Gyawali**, Matei Budiu, Tej Chajed, Frank McSherry, Val Tannen. **DBSP: Automatic Incremental View Maintenance for Rich Query Languages**. *The VLDB Journal*, vol. 34, 2025. (“Best of VLDB 2023” special issue.)

RESEARCH & INDUSTRY EXPERIENCE

Feldera | Software Engineer US, Remote
Dec 2023 – July 2026

- Designed **fault tolerance** for long-running incremental pipelines: automatic **S3 checkpoint synchronization** with multihost coordination, so pipelines recover from crashes and migrate across machines without recomputing from scratch.
- Built **change-data-capture** connectors that integrate incrementally maintained views with external systems: **Postgres** (CDC input and output, TLS) and **Redis** (transactional output).
- Applied **differential testing** to the SQL-to-DBSP compiler by extending **SQLancer** with a Feldera backend (NoREC oracle), uncovering **31 previously undetected bugs**.
- Extended the compiler with built-ins for array operations, temporal arithmetic, and numeric rounding.
- Maintain the **Feldera Python SDK** and much of the user-facing documentation.

Invisid | Software Developer Germany, Remote
May 2022 – Nov 2023

- Developed **trackpad-rs**, the input-capture layer for continuous biometric authentication on macOS.
- Redesigned the behavioral-authentication backend, improving efficiency and consistency of biometric data processing.

Flow Webinar | Security Engineer (Part-time) US, Remote
May 2020 – June 2021

- Performed vulnerability assessment and penetration testing of a WebRTC video conferencing platform; administered production backend infrastructure.

FELLOWSHIPS & AWARDS

UTCS Distinguished Graduate Fellowship | University of Texas at Austin 2026

DWIT Merit-Based Scholarship | Deerwalk Institute of Technology 2020, 2021

PROJECTS

Feldera [Rust](#) [SQL](#) [Python](#) Incremental computation on DBSP [1]; 340+ commits. [GitHub](#) [feldera/feldera](#)

SocketDB [Rust](#) [SQL](#) SQL database streaming live query updates over WebSockets. [GitHub](#) [abhizer/socketdb](#)

trackpad-rs [Rust](#) Exposes raw trackpad and mouse data from Apple’s *MultitouchSupport*. [GitHub](#) [trackpad-rs](#)

Nyx-lang [Rust](#) Statically typed, tree-walking interpreted language with full type checking. [GitHub](#) [abhizer/nyx-lang](#)

Loogle-rs [Rust](#) Local full-text search engine based on TF-IDF ranking. [GitHub](#) [abhizer/loogle-rs](#)